Color Picker Documentation

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Overview

System Summary:

This is a simple Unity Color picker that supports user selected coloring and code-access to color features and details. It's made of a single script and a color picker prefab. It allows custom UI designs, and you can disable certain features simply by deleting their graphics from the hierarchy. It is made to be modular and can be pasted into any project.

Why I made it:

I wanted to make a color picker which is generic and modular so that I don't have to spend time making a color picker every time I needed one. I specifically made it with plans to use it in my P5 Designer project.

Features:

- User inputs colors by editing RGBA values separately or using hue bar and saturation/brightness box.
- All features (excluding the hue bar and saturation/brightness box) can be deleted safely.
- Automatically displays previous colors (can be easily changed manually through code).
- Dynamically adapts colors to match current RGBA values
- Includes performance settings

Installation & Setup

1. Use the provided Unity Package to quickly import scripts into your project.
2. Drag and drop the Color Picker prefab into your canvas.
3. Assign the “Canvas” and “Raycaster” variables using the Canvas component of your canvas and the Graphic Raycaster component of your canvas respectively.
4. Adjust the prefab to match your UI design
5. If your Hue slider is horizontal, you will need to select Horizontal for the H Slider Mode variable under Hue Settings in the Color Picker component.

NOTE: If you wish to set the previous color manually, you will need to remove the OnEnable function entirely from the ColorPicker.cs file, so that the color picker stops setting it automatically.

Script Breakdown

ColorPicker.cs:

Contains all the code related to the color picker.

Public Properties:

- **color CurrentColor:** Returns current color of the color picker
- **float R:** Returns the red value of the current color (from 0 to 1)
- **float G:** Returns the green value of the current color (from 0 to 1)
- **float B:** Returns the blue value of the current color (from 0 to 1)
- **float A:** Returns the alpha value of the current color (from 0 to 1)
- **float Hue:** Returns the hue value of the current color (from 0 to 1)
- **float Saturation:** Returns the saturation value of the current color (from 0 to 1)
- **float Brightness:** Returns the brightness value of the current color (from 0 to 1)
- **string Hex:** Returns the hexadecimal code of the current color

Public Methods:

- **SetPreviousColor(Color color):**
  - Sets the color of the Previous Color Image of the color picker to **color**.
  - Does nothing if the previous Color Image is null (deleted from scene).
- **SetColor(Color color):**
  - Sets current color to **color**.
- **SetColorHSV(float h, float s, float v):**
  - Sets hue, saturation, and value (brightness) of the current color.
- **SetColorHex(string hexVal):**
  - Sets current color to the hexadecimal value **hexVal**.
- **SetHue(float val):**
  - Sets hue of the current color.
- **SetSaturation(float val):**
  - Sets saturation of the current color.
- **SetBrightness(float val):**
  - Sets brightness/value of the current color.
- **SetR(float val):**
  - Sets red value of the current color.
- **SetG(float val):**
  - Sets green value of the current color.
- **SetB(float val):**
  - Sets blue value of the current color.
- **SetAlpha(float val):**
  - Sets alpha value of the current color.

Events:

- **OnColorChanged(Color color):** Invokes when current color changes.

Code Examples

- Accessing Current Color

  ```
  // Setting a color to the current color on the color picker

  ColorPicker picker;

  Color myColor = picker.CurrentColor;
  ```

- Setting Previous Color

  ```
  /* Note that the color picker automatically sets the previous color,

  so if you want to do it manually, you must delete the OnEnable function in the ColorPicker class
  */

  ColorPicker picker;

  Color previousColor = Color.white;

  picker.SetPreviousColor(previousColor);
  ```